

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026

Chemistry — Class IX | Paper: English Medium

Syllabus: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)

Total Marks: 65

Time: 2h 30m

Version: 2

Atomic Masses (amu): H=1 | C=12 | N=14 | O=16 | Na=23 | Mg=24 | S=32 | Cl=35.5 | K=39 | Ca=40 | Cr=52 | Fe=56 | Cu=63.5 | Zn=65 Avogadro's Number: 6.022×10^{23}

General Instructions:

- Write your Roll Number in the space provided on the answer sheet. Do not write your name.
- Attempt all questions in Section A. In Sections B and C, attempt each question OR its alternative, not both.
- Use blue or black ink pen. Pencil may be used for diagrams only.
- All working must be shown for Section B and C questions. Marks will be deducted for incomplete working.

SECTION A — Multiple Choice Questions ($12 \times 1 = 12$ Marks)

#	Question	A	B	C	D	Answer
1	Which branch of chemistry studies physical and chemical changes in living organisms?	Ⓐ Medicinal	Ⓑ Biochemistry	Ⓒ Analytical	Ⓓ Nuclear	ⒶⒷⒸⒹ
2	Which state of matter is electrically conductive and found in the sun and lightning?	Ⓐ Liquid crystal	Ⓑ BEC	Ⓒ Plasma	Ⓓ Gel	ⒶⒷⒸⒹ
3	Which was a major defect of Rutherford's atomic model?	Ⓐ It denied the nucleus	Ⓑ Electrons spiral into nucleus, emitting a continuous spectrum	Ⓒ It could not explain isotopes	Ⓓ It proposed quantized orbits	ⒶⒷⒸⒹ
4	An atom has atomic number 17 and mass number 35. How many neutrons does it have?	Ⓐ 17	Ⓑ 35	Ⓒ 18	Ⓓ 52	ⒶⒷⒸⒹ
5	Which element has the highest electronegativity on the Pauling scale?	Ⓐ Cl	Ⓑ O	Ⓒ F	Ⓓ N	ⒶⒷⒸⒹ
6	Which halogen exists as a red-brown liquid at room temperature?	Ⓐ Fluorine	Ⓑ Chlorine	Ⓒ Bromine	Ⓓ Iodine	ⒶⒷⒸⒹ
7	In a coordinate covalent bond, both bonding electrons are donated by:	Ⓐ Both atoms equally	Ⓑ The more electronegative atom	Ⓒ One atom (the donor)	Ⓓ The receptor atom	ⒶⒷⒸⒹ
8	What is the formula mass of KCl?	Ⓐ 58.5 amu	Ⓑ 74.5 amu	Ⓒ 63 amu	Ⓓ 82 amu	ⒶⒷⒸⒹ
9	How many molecules are present in 0.5 mol of CH_4 ?	Ⓐ 6.022×10^{23}	Ⓑ 3.011×10^{23}	Ⓒ 1.204×10^{24}	Ⓓ 1.5×10^{23}	ⒶⒷⒸⒹ
10	In $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$, which substance is the oxidizing agent?	Ⓐ Na	Ⓑ Cl_2	Ⓒ NaCl	Ⓓ Both Na and Cl_2	ⒶⒷⒸⒹ
11	A catalyst increases the rate of a reaction by:	Ⓐ Increasing enthalpy change	Ⓑ Lowering activation energy	Ⓒ Increasing activation energy	Ⓓ Consuming itself	ⒶⒷⒸⒹ
12	At chemical equilibrium, which statement is true?	Ⓐ Forward reaction stops	Ⓑ Reverse reaction stops	Ⓒ Both reactions stop	Ⓓ Both reactions continue at the same rate	ⒶⒷⒸⒹ

SECTION B — Short Questions ($11 \times 3 = 33$ Marks)

Attempt all 11 questions. Choose either the main question OR its alternative, not both.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Differentiate between industrial chemistry and nuclear chemistry. Give one example of each.	3	OR	Explain the difference between science, technology, and engineering with one example each.	3
2(ii)	Compare the macroscopic properties (density, compressibility, flow) of solids, liquids, and gases in tabular form.	3	OR	What are allotropes? Describe the structural differences between graphite and diamond and explain why graphite conducts electricity but diamond does not.	3
2(iii)	State any THREE postulates of Bohr's atomic model. How did Bohr explain the emission of light by atoms?	3	OR	Define isotopes. Name the three isotopes of hydrogen, give their mass numbers, and identify	3

				which one is radioactive.	
2(iv)	Write the electronic configuration of Na (Z=11) and Al (Z=13). State the group and period of each in the periodic table.	3	OR	Explain the trend of ionization energy across a period and down a group with reference to nuclear charge and shielding effect.	3
2(v)	Define ionic bond. Explain, using CaCl_2 as an example, how the formula of an ionic compound is derived from the charges on its ions.	3	OR	Distinguish between polar and non-polar covalent bonds with two examples of each. What determines bond polarity?	3
2(vi)	Find the empirical formula of a compound containing 40% C, 6.7% H, and 53.3% O by mass. (C=12, H=1, O=16)	3	OR	Calculate the mass of 2.5 mol of sucrose $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ and the number of molecules present. (M of sucrose = 342 g/mol)	3
2(vii)	Balance the equation: $\text{Al} + \text{O}_2 \rightarrow \text{Al}_2\text{O}_3$. Identify the type of reaction.	3	OR	Calculate the number of moles in 100 g of H_2O and in 5 g of H_2 . (H=1, O=16)	3
2(viii)	State FOUR rules for assigning oxidation states. Determine the oxidation state of Cr in $\text{K}_2\text{Cr}_2\text{O}_7$.	3	OR	Define oxidizing and reducing agents. In $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$, identify each agent and give a reason.	3
2(ix)	What is corrosion? Describe TWO methods used to prevent corrosion of iron, giving the principle behind each.	3	OR	When acidified KMnO_4 is added to a reducing agent, what colour change occurs? Explain. What is the change in oxidation number of Mn?	3
2(x)	Define enthalpy of reaction. State the sign of ΔH for exothermic and endothermic reactions. Give one thermochemical equation for each.	3	OR	Define activation energy. Explain how a catalyst changes activation energy, with reference to the energy pathway diagram.	3
2(xi)	What is a reversible reaction? Write the equation for the dehydration of hydrated copper(II) sulphate and state the colour changes on heating and on adding water.	3	OR	State THREE conditions required for a chemical system to maintain equilibrium. What is meant by dynamic equilibrium?	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	(a) Describe Dalton's atomic theory: state THREE postulates and mention ONE limitation. (3) (b) An element has Z=19 and A=39. Find protons, neutrons, and electrons. Write its electronic configuration; give its group and period. (2)	5	OR	(a) Explain how the periodic table is organized. Define s-block, p-block, and d-block with one example each. (3) (b) Explain how atomic size changes (i) across a period and (ii) down a group, giving a reason for each trend. (2)	5
3(ii)	(a) Define mole and Avogadro's number. Calculate the mass of 9.05 mol of ozone O_3 . (O=16) (3) (b) Write the balanced net ionic equation for Mg reacting with dilute H_2SO_4 . Identify the spectator ions. (2)	5	OR	(a) Distinguish between empirical and molecular formula. The empirical formula of benzene is CH and its molecular mass is 78 amu. Find its molecular formula. (3) (b) Find the number of atoms in 1.25 mol of Zn and in 0.2 mol of Al. ($N_A = 6.022 \times 10^{23}$) (2)	5
3(iii)	(a) Define redox reactions in terms of (i) oxygen transfer, (ii) hydrogen transfer, (iii) electron transfer, with one example each. (3) (b) Determine the oxidation state of S in H_2SO_4 and N in HNO_3 . Show all working. (2)	5	OR	(a) Describe cathodic (sacrificial) protection as a corrosion prevention method. State the principle and give one application. (3) (b) For the reaction $\text{CuO} + \text{Zn} \rightarrow \text{Cu} + \text{ZnO}$, identify which element is oxidized and which is reduced. Name the oxidizing and reducing agents. (2)	5
3(iv)	(a) Calculate ΔH for $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ using bond energies: H-H = 436, Cl-Cl = 243, H-Cl = 432 kJ/mol. State whether the reaction is exo- or endothermic. (3) (b) What is aerobic respiration? Write the word equation and explain why it is an exothermic process. (2)	5	OR	(a) Explain the effect of temperature on the hydrated cobalt(II) chloride equilibrium. Write the equation and describe the colour changes on heating and on adding water. (3) (b) Define system and surroundings. Give one exothermic and one endothermic thermochemical equation with their ΔH values. (2)	5

Invigilator's Signature

Candidate's Signature / Roll No.

Chapters Covered: Ch 1 (Nature of Chemistry) | Ch 2 (Matter) | Ch 3 (Atomic Structure) | Ch 4 (Periodic Table) | Ch 5 (Chemical Bonding) | Ch 6 (Stoichiometry) | Ch 7 (Electrochemistry) | Ch 8 (Energetics) | Ch 9 (Chemical Equilibrium)

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